(You may use the back side of the sheet to write down your answers.)

Question 1 (15 points)

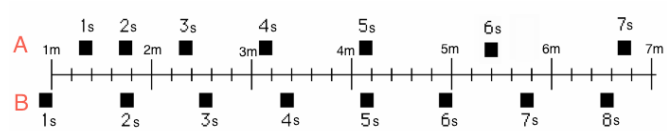
Read the statements below and decide whether they are true or false. Explain. Give at least one example supporting your answer. (No credit will be given to an answer without an explanation.)

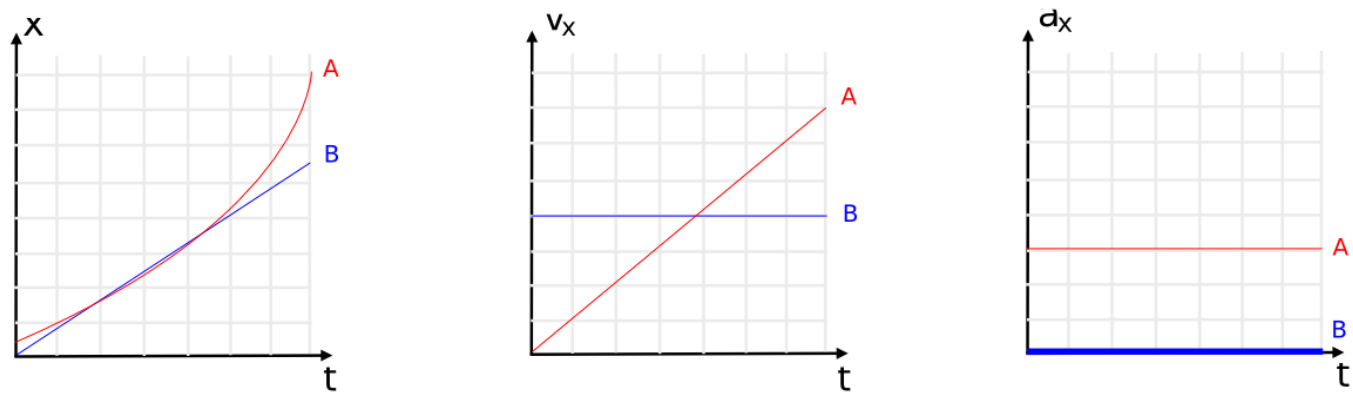
(i) If an object is in equilibrium, then it must be at rest.
If an object is in equilibrium then there is no net force acting the object ($\sum F = 0$). According to Newton's 1st law; if there is no net force acting on the object, the object is either at rest or if it is moving initially it will keep moving with the same initial velocity.

(ii) I can move with constant speed and still accelerate.
Consider the uniform circular motion, where the speed of the object constant. Since its velocity vector continuously changes direction there will be a radial acceleration due to the change in the direction of the velocity vector.

Question 2 (10 points)

(i) The x-positions of two blocks at successive 1.0-second time intervals are shown in the figure. Draw x vs. t, v_x vs. t and a_x vs. t graphs for both blocks. Show block A and block B on the same graph.



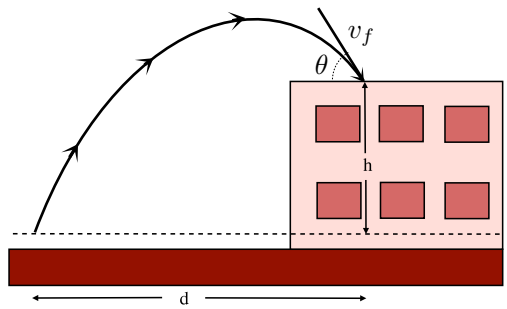


(ii) What is the average velocity of block A between t =1 s and t =6 s?

$$\boxed{v_{\text{avg}} = \frac{x_2 - x_1}{t_2 - t_1}} \quad \Rightarrow \quad v_{\text{avg}} = \frac{5.4\text{m} - 1.4\text{m}}{6.0\text{s} - 1.0\text{s}} = 0.80 \text{ m/s}$$

Problem (1) (25 points)

A ball is thrown up onto a roof, as shown in the figure. It lands at height $h = 21.0$ m above where it is released. Just before landing, the ball's path is angled at $\theta = 53.1°$ with the roof and its speed is $v_f = 10.0$ m/s. (Take $g = 10$ m/s^2; $\sin 53.1° = 0.8$, $\cos 53.1° = 0.6$.)



(a) (15pts.) Find the horizontal distance it travels.

You can think of the situation as the ball being fired from the roof of a building with the given initial velocity. When the ball lands on the ground it travels a horizontal distance d from where it is fired. You need to find the time that it takes the ball from roof to the ground, apply the following formula and solve the quadratic equation to find the time;

$$\boxed{y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2} \quad \Rightarrow \quad 0 = (21.0\text{m}) + (10.0\text{m/s})\sin 53.1^o\ t + \frac{1}{2}(-10.0\text{m/s}^2)\ t^2$$

$$5t^2 - 8t - 21 = 0 \quad \Rightarrow \quad (5t+7)(t-3) = 0 \quad \Rightarrow \quad t = 3.0\text{s}$$

On horizontal axis, the ball travels with a constant velocity which is equal to the horizontal component of initial velocity, therefore;

$$\boxed{d = v_{0x}t} \quad \Rightarrow \quad d = (10.0\text{m/s})\cos 53.1^o(3.0\text{s}) \quad \Rightarrow \quad \boxed{d = 18.0 \text{ m}}$$

(b) (10pts.) Find the initial velocity of the ball in unit-vector notation.

Horizontal component of velocity is constant during the motion. Final vertical component of its velocity can be found by the following formula;

$$\boxed{v_y = v_{0y} + a_y t} \quad \Rightarrow \quad v_y = (10.0\text{m/s})\sin 53.1^o + (-10.0\text{m/s}^2)(3.0\text{s}) \quad \Rightarrow \quad v_y = -22.0 \text{ m/s}$$

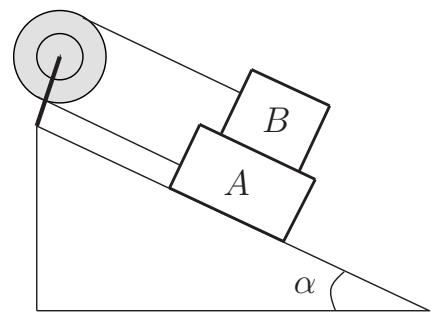
This is the y component of the inverse trajectory. In our case its fired upwards so: $v_{initial_y} = 22.0\text{m/s}$

$$v_x = (10.0\text{m/s})\cos 53.1^o \quad \Rightarrow \quad v_x = 6.0 \text{ m/s}$$

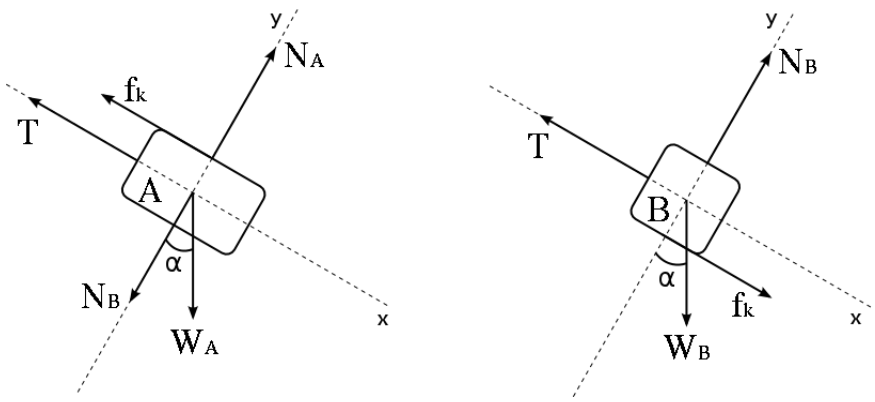
$$\boxed{\vec{v_i} = v_x\hat{\imath} + v_y\hat{\jmath}} \quad \Rightarrow \quad \boxed{\vec{v_i} = (6.0 \text{ m/s})\hat{\imath} + (22.0 \text{ m/s})\hat{\jmath}}$$

Problem (2) (25 points)

Two blocks with weights $w_A = 20$ N and $w_B = 16$ N, connected by a light rope passing through a frictionless pulley, are placed on an inclined surface ($\alpha = 30.0^0$) as shown in the figure. The block A moves with a constant velocity in downward direction, and there is no friction between block A and the surface.



(a) (15pts.) Draw free-body diagrams for each block.



(b) (10pts.) Find the magnitude of the friction force between the blocks.

Since the block A moves with a constant velocity, block B also moves with the same constant velocity. The acceleration of the blocks is zero. Apply Newton's 2nd law for the blocks both in horizontal (x) axis;

for block A:

$$\sum F_x = m_A\ a_x$$

$$\boxed{W_A \sin\alpha - f_k - T = 0}$$

for block B:

$$\sum F_x = m_B\ a_x$$

$$W_B \sin\alpha + f_k - T = 0$$

$$\boxed{T = W_B \sin\alpha + f_k}$$

$$W_A \sin\alpha - f_k - (W_B \sin\alpha + f_k) = 0$$

$$\boxed{f_k = \frac{(W_A - W_B)\sin\alpha}{2}} \quad \Rightarrow \quad f_k = \frac{(20.0\text{N} - 16.0\text{N})\sin 30.0^o}{2} \quad \Rightarrow \quad \boxed{f_k = 1.0\ \text{N}}$$

Problem (3) (25 points)

The radius of the (nearly circular) Nişantepe roundabout is 43.2 m. Your car has mass 1500 kg (including the driver). The coefficient of static friction between the asphalt and the car tire is 0.75. (Take $g = 10$ m/s^2.)

(a) (10pts.) What is the maximum speed you can have to drive through the Nişantepe roundabout without sliding (in km/h)?

The only force acting on the car is the friction force, which is radial inwards direction and equals to: $\boxed{F_f = \mu_s N = \mu_s m_{\text{car}}\ g}$. Then Newtons 2nd law states that:

$\boxed{m_{\text{car}}\ a_{\text{radial}} = F_{\text{total}} = F_f = \mu_s m_{\text{car}}\ g}$. And we know that $\boxed{a_{\text{radial}} = \dfrac{v^2}{R}} \rightarrow \boxed{\dfrac{v^2}{R} = \mu_s g} \rightarrow$

$\boxed{v = \sqrt{\mu_s g R}} \rightarrow v = \sqrt{0.75 \times 43.2 \times 10}\ \text{m/s} = \sqrt{324}\ \text{m/s} = 18\ \text{m/s}.$

$18\text{m/s} = 18 \times \dfrac{3600}{1000}\text{km/h} \rightarrow \boxed{v_{\text{max}} = 64.8\text{km/h}}$

(b) (8pts.) Two of your friends say to you: "More passengers mean more mass. If you give us a lift every morning, you will drive the Nişantepe roundabout more safely". How should you answer?

It is obvious from the solution above that v_{max} is independent of the mass of the car so the passengers. More passengers will not effect the maximum speed.

(c) (7pts.) While driving in the roundabout together with your friends, they say: "We are pushed outward because there is a centrifugal force acting on us". How should you react?

"There is NO centrifugal force!" in this motion. They were not pushed outwards, they only wanted to preserve their velocity. In fact they were pulled inwards by a CENTRIPETAL force applied ON them by the car (or applied on the car by friction). Due to this force they are passing the roundabout with constant speed but NOT constant velocity.

Formula Sheet

Scalar Product:

$c = \vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$

$c = AB \cos\phi$

Vector Product:

$\vec{C} = \vec{A} \times \vec{B}$

$C = AB \sin\phi$

$C_x = A_y B_z - A_z B_y$

$C_y = A_z B_x - A_x B_z$

$C_z = A_x B_y - A_y B_x$

Straight-line motion:

$$v_{av-x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}}$$

$$v_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta x}}{\boldsymbol{\Delta t}} = \frac{dx}{dt}$$

$$a_{av-x} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} = \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}}$$

$$a_x = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta v_x}}{\boldsymbol{\Delta t}} = \frac{dv_x}{dt}$$

Constant x-acceleration:

$v_x = v_{0x} + a_x t$

$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$

$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$

$$x - x_0 = \left(\frac{v_{0x} + v_x}{2}\right) t$$

Motion in three dimensions:

$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}$

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}}$$

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{r}}}{\boldsymbol{\Delta t}} = \frac{d\vec{r}}{dt}$$

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}}$$

$$\vec{a} = \lim_{\Delta t \to 0} \frac{\boldsymbol{\Delta \vec{v}}}{\boldsymbol{\Delta t}} = \frac{d\vec{v}}{dt}$$

Circular motion:

$$a_{\text{rad}} = \frac{v^2}{R} \quad a_{\text{rad}} = \frac{4\pi^2 R}{T^2}$$

Relative velocity:

$v_{\text{P/A-}x} = v_{\text{P/B-}x} + v_{\text{B/A-}x}$

$\vec{v}_{\text{P/A}} = \vec{v}_{\text{P/B}} + \vec{v}_{\text{B/A}}$

Newton's 1st Law:

$\vec{R} = \vec{F}_1 + \vec{F}_2 + \ldots = \sum \vec{F}$

$\sum \vec{F} = 0$

Newton's 2nd Law:

$\sum \vec{F} = m\vec{a}$

Newton's 3rd Law:

$\vec{F}_{\text{A on B}} = -\vec{F}_{\text{B on A}}$

Friction Force:

$f_k = \mu_k n \quad f_s \leq \mu_s n$

Work done by a force:

$W = \vec{F} \cdot \vec{s} = Fs \cos\phi$

Kinetic energy:

$K = \frac{1}{2} m v^2$

The work-energy theorem:

$W_{\text{tot}} = K_2 - K1 = \Delta K$

Work done by a varying force:

$$W = \int_{x_1}^{x_2} F_x \, dx$$

Power:

$$P_{\text{av}} = \frac{\Delta W}{\Delta t}$$

$P_{\text{av}} = \vec{F} \cdot \vec{v}$

Note: If you detach this page during the exam, return it at the end with your exam sheet after you have signed below.

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